

NENANA, AK, USA

WMO: 702600

Lat: 64.550N

Lon: 149.072W

Elev: 360

StdP: 14.51

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26435

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.9	-37.6	-43.5	0.4	-18.0	-35.3	0.7	-21.3	26.1	3.7	23.1	7.8	1.2	100	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	80.6	61.0	76.9	59.9	73.5	58.3	63.8	76.5	61.9	73.3	60.1	70.2	5.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	75.5	66.4	57.1	70.6	64.6	55.4	66.3	63.3	29.1	76.6	27.8	73.2	26.6	69.9	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.2	15.1	-46.7	85.3	6.1	3.3	-51.1	87.7	-54.7	89.7	-58.1	91.5	-62.5	93.9		
(5)			-34.2	67.2	4.0	2.5	-37.1	69.0	-39.4	70.4	-41.7	71.8	-44.6	73.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	-3.3	4.6	10.7	32.5	49.1	59.2	61.4	56.5	45.8	27.5	6.4	-1.1		
	DBStd	26.82	19.38	16.85	14.77	10.62	7.80	5.27	4.64	6.07	7.34	11.11	14.92	16.75		
	HDD50	8541	1652	1272	1218	525	109	2	1	13	160	697	1308	1584		
	HDD65	13088	2117	1692	1683	974	493	185	129	272	575	1161	1758	2049		
	CDD50	964	0	0	0	1	82	279	353	214	35	0	0	0		
	CDD65	37	0	0	0	0	1	12	16	8	0	0	0	0		
	CDH74	654	0	0	0	0	53	230	274	97	0	0	0	0		
	CDH80	99	0	0	0	0	6	36	46	11	0	0	0	0		
	WSAvg	5.2	5.4	5.5	5.6	5.9	5.5	4.8	4.4	4.3	5.0	5.3	5.4	5.4		
	PrecAvg	11.9	0.5	0.4	0.3	0.3	0.6	1.6	3.3	2.6	1.6	0.9	0.8	0.5		
(15) Precipitation	PrecMax	16.7	2.1	1.7	2.5	3.1	2.8	4.8	7.6	5.4	6.7	5.0	2.1	1.5		
	PrecMin	7.6	0.0	0.0	0.0	0.0	0.1	0.2	1.3	0.8	0.2	0.2	0.0	0.1		
	PrecStd	2.4	0.5	0.4	0.6	0.7	0.6	1.2	1.9	1.3	1.3	1.0	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	44.4	46.4	64.3	80.3	84.1	84.6	81.3	68.1	56.1	38.4	38.3		
		MCWB	36.7	36.5	37.7	47.4	55.5	62.0	64.2	62.7	54.5	45.7	33.9	33.5		
	2%	DB	35.1	36.7	41.4	57.5	73.7	80.2	80.5	76.7	64.2	49.6	34.3	32.5		
		MCWB	30.8	31.0	35.1	43.2	53.9	60.2	62.2	61.6	52.1	42.3	30.9	28.9		
	5%	DB	29.8	33.2	36.7	53.6	69.5	76.1	76.8	71.9	61.2	45.5	30.8	27.9		
		MCWB	26.3	29.1	30.7	40.9	51.8	58.7	60.7	59.3	50.7	39.6	28.2	24.6		
	10%	DB	24.6	28.3	32.0	49.0	65.3	72.8	73.5	67.8	57.8	41.6	26.8	21.7		
		MCWB	21.8	24.8	27.4	38.9	49.4	57.0	59.5	57.2	49.9	36.9	24.2	19.4		
	0.4%	WB	37.1	36.9	39.9	48.5	57.2	65.6	66.5	66.0	56.9	47.5	34.4	34.2		
		MCDB	42.4	43.5	45.1	62.1	74.7	79.5	79.9	77.2	62.2	53.9	37.0	38.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.3	32.4	34.8	44.2	54.9	62.5	64.0	62.9	54.3	43.0	31.6	29.5		
		MCDB	35.4	36.4	41.0	55.4	71.2	75.8	76.8	74.0	60.5	48.7	34.1	32.8		
	5%	WB	26.9	28.9	31.1	41.7	53.0	60.2	62.3	60.4	52.5	40.0	28.3	25.3		
		MCDB	30.3	33.2	36.5	51.8	66.7	72.5	73.9	69.0	58.9	44.8	31.0	28.3		
	10%	WB	22.8	25.1	27.5	39.4	50.9	58.5	60.7	58.6	50.7	37.2	24.2	20.3		
		MCDB	25.4	28.9	32.2	49.0	63.1	70.0	71.1	65.7	57.2	41.5	26.7	22.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	20.0	23.3	22.2	23.1	22.0	19.8	19.2	18.9	14.2	14.3	16.8		
		MCDBR	18.9	20.1	23.5	27.7	30.1	28.0	26.9	25.4	23.3	18.9	16.3	22.2		
		MCWBR	16.0	16.0	18.6	16.4	13.7	12.1	12.8	13.7	14.6	14.4	13.9	19.6		
		MCDBR	19.2	19.7	22.6	25.9	26.4	25.1	23.8	22.1	20.7	18.5	16.1	22.3		
	5% WB	MCWBR	16.5	16.0	18.3	16.6	12.4	11.9	12.4	13.1	14.6	14.5	14.0	19.7		
(40) Clear-Sky Solar Irradiance	taub		0.177	0.249	0.255	0.307	0.347	0.384	0.379	0.392	0.312	0.267	N/A	0.179		
	taud		1.950	2.205	2.303	2.297	2.421	2.331	2.368	2.330	2.538	2.463	N/A	1.939		
	Ebn at Noon		124	213	270	275	274	263	261	244	249	213	N/A	43		
	Edh at Noon		8	17	24	32	31	35	33	31	20	14	N/A	3		
(44) All-Sky Solar Radiation	RadAvg		54	278	849	1459	1678	1783	1536	1174	741	300	79	17		
	RadStd		4	21	52	97	123	112	106	108	95	32	12	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.36	N/A	N/A	N/A	N/A	N/A	-1118	-1228	N/A	N/A
(47) Regional (7 neighbors)		+1.67	N/A	N/A	N/A	N/A	N/A	-599	-682	N/A	N/A

Nomenclature: See separate page